Newton's First Law

Laws of Motion + Friction · Class 11

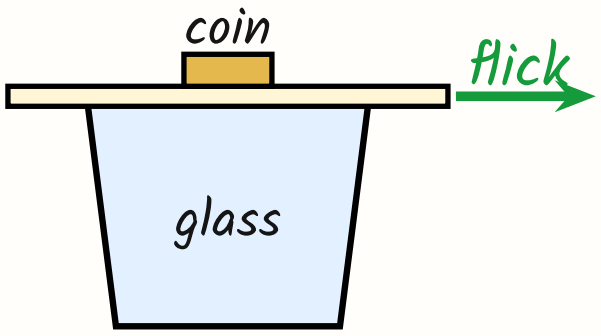
1 Law of inertia

$$\vec{F}_{net} = 0 \iff \vec{v} = \text{constant}$$

$\vec{F}_{net}$: sum of all EXTERNAL forces · rest ($\vec{v} = 0$) included · INERTIAL frame

- Inertial frame: not accelerating, not rotating. The ground is nearly one (Earth's spin neglected).

2 Inertia



- Inertia: resistance to a change of velocity. Mass is its measure.
- Card flicked fast: friction acts on the coin too briefly to move it; the coin drops in.

Golden points

★ A net force is needed to CHANGE velocity, not to keep it.

★ Car at constant velocity on a level road: $F_{net} = 0$ (road friction on the driving wheels balances drag).

★ $F_{net} = 0$ does not mean no forces: a book at rest on a horizontal table (no other vertical force) has $N = mg$.

★ Bus brakes: the passenger's upper body keeps its velocity and leans forward (ground frame).

Summary

$F_{net} = 0 \Leftrightarrow \vec{v}$ constant • inertia $\propto$ mass • inertial frame

Momentum and the Second Law

1 Linear momentum

$$\vec{p} = m\vec{v}$$

along $\vec{v}$ · $\mathrm{kg\,m\,s^{-1}} = \mathrm{N\,s}$ · $[\mathrm{MLT^{-1}}]$

2 Second law

$$\vec{F}_{net} = \frac{d\vec{p}}{dt} = m\vec{a} \quad (\text{constant } m)$$

$\vec{F}_{net}$: net EXTERNAL force · inertial frame · $1\,\mathrm{N} = 1\,\mathrm{kg\,m\,s^{-2}}$

- $\vec{a}$ is along $\vec{F}_{net}$, not necessarily along $\vec{v}$.
- Per axis: $F_x = ma_x$, $F_y = ma_y$; $\vec{a}$ at each instant is set by $\vec{F}_{net}$ at that instant.

Example

$m = 2\,\mathrm{kg}$, only force $\vec{F} = (6\hat{i} + 8\hat{j})\,\mathrm{N}$: $\vec{a} = (3\hat{i} + 4\hat{j})\,\mathrm{m\,s^{-2}}$, $|\vec{a}| = 5\,\mathrm{m\,s^{-2}}$.

3 Changing mass: sand on a belt

$$F = v\frac{dm}{dt}$$

sand falls vertically onto a belt at constant v · dm/dt: kg added per second · F: extra force to keep v

Example

$dm/dt = 2\,\mathrm{kg\,s^{-1}}$, $v = 3\,\mathrm{m\,s^{-1}}$: $F = 6\,\mathrm{N}$.

Impulse

1 Impulse-momentum theorem

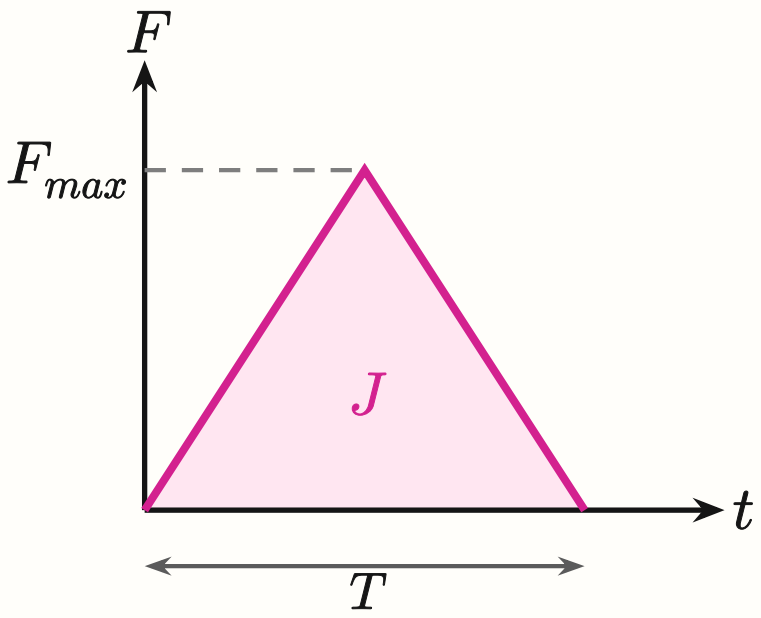
$$\vec{J} = \int \vec{F}\,dt = \Delta\vec{p}, \qquad \vec{F}_{avg} = \frac{\Delta\vec{p}}{\Delta t}$$

$\vec{F}$: NET force on the body · $\vec{J}$ along $\Delta\vec{p}$ · unit $\mathrm{N\,s}$

Example

$0.2\,\mathrm{kg}$ ball hits a wall normally at $10\,\mathrm{m\,s^{-1}}$, rebounds at $10\,\mathrm{m\,s^{-1}}$ in $0.01\,\mathrm{s}$: $|\Delta p| = 4\,\mathrm{N\,s}$, $F_{avg} = 400\,\mathrm{N}$ on it, away from the wall.

2 Area under the F–t graph



- Impulse = area under the F–t graph.
- Triangle: $J = \frac{1}{2}F_{max}T$.
- $F_{max} = 600\,\mathrm{N}$, $T = 0.02\,\mathrm{s}$: $J = 6\,\mathrm{N\,s}$; a $0.3\,\mathrm{kg}$ ball at rest leaves at $20\,\mathrm{m\,s^{-1}}$.

Golden points

★ Impulsive force: huge and brief (bat on ball); in that time, gravity's impulse is negligible.

★ Same Δp over a longer Δt: smaller average force (hands pulled back while catching, airbags, sand pits).

★ Head-on rebound (velocity reverses along the same line): $|\Delta p| = m(u + v)$.

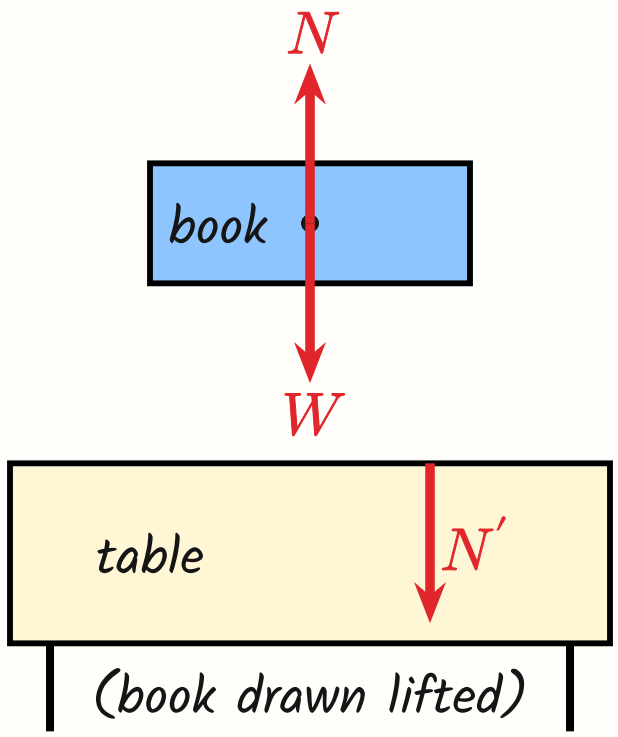
Newton's Third Law

1 Action and reaction

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

$\vec{F}_{AB}$: ON A BY B · on DIFFERENT bodies · same instant · same type

2 Book on a table



- On the book: N (by the table) and W (by the Earth). $N = W$ (book at rest on a horizontal table), but NOT a pair: both act on the book.
- Partner of N: N' on the table, down.
- Partner of W: book pulls the Earth up.

3 Pairs never cancel

- They act on different bodies; a body moves by the forces ON it.
- Horse and cart: ground friction on the hooves (external, forward) beats the resistance on the cart (while they speed up; at constant velocity the two are equal).

Example

Earth pulls a $1\,\mathrm{kg}$ apple with $10\,\mathrm{N}$ ($g = 10\,\mathrm{m\,s^{-2}}$ near Earth's surface); the apple pulls Earth ($6 \times 10^{24}\,\mathrm{kg}$) with $10\,\mathrm{N}$: $a_{Earth} \approx 1.7 \times 10^{-24}\,\mathrm{m\,s^{-2}}$.

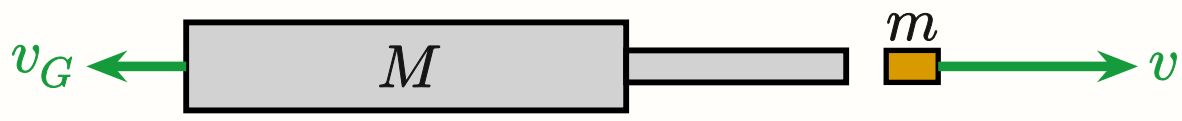
Conservation of Momentum

1 Isolated system

$$\sum \vec{F}_{ext} = 0 \;\Rightarrow\; \vec{P} = \sum m_i \vec{v}_i = \text{constant}$$

internal forces cancel in pairs (3rd law) · per axis: $F_{ext,x} = 0 \Rightarrow P_x$ constant

2 Recoil of a gun



(speeds not to scale)

$$0 = mv + Mv_G \;\Rightarrow\; v_G = -\frac{m\,v}{M}$$

both at rest before · gun free to recoil (no horizontal external force) · ground frame

Example

$M = 4\,\text{kg}$, $m = 20\,\text{g}$, $v = 400\,\text{m s}^{-1}$: $v_G = -2\,\text{m s}^{-1}$. *KE: bullet* $1600\,\text{J}$, *gun* $8\,\text{J}$ *(ratio* $M/m = 200$*).*

3 Explosion of a body at rest

- *Two pieces get equal and opposite momenta (no external impulse during the burst):* $m_1 v_1 = m_2 v_2$. *A* $3\,\text{kg}$ *shell bursts:* $1\,\text{kg}$ *at* $12\,\text{m s}^{-1}$, $2\,\text{kg}$ *at* $6\,\text{m s}^{-1}$ *the other way.*

Golden points

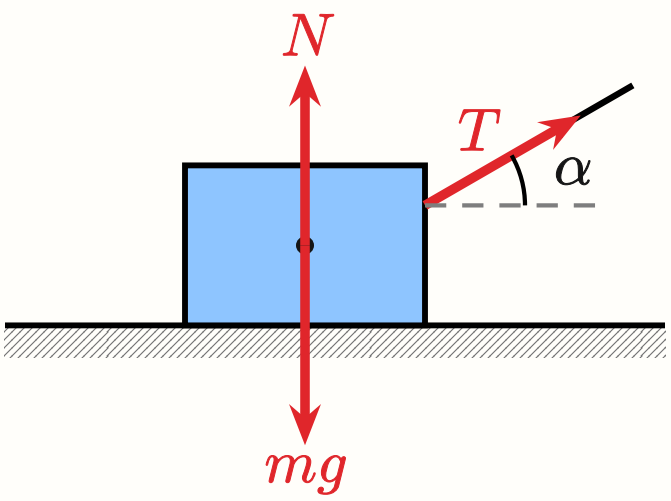
★ *KE is NOT conserved in explosions and inelastic collisions; momentum is (external impulse negligible during the event).*

Free Body Diagrams

1 Rules

- ONE body; only forces ON it, each with an agent; no "ma" arrow (inertial frame).
- mg: vertically down (g uniform near Earth's surface).
- N: $\perp$ the surface, a push; $N = 0$ when contact is lost.
- T: along the string, away from the body. f: along the surface, against slipping (or its tendency).

2 String at angle α, smooth floor



$$N = mg - T\sin\alpha$$

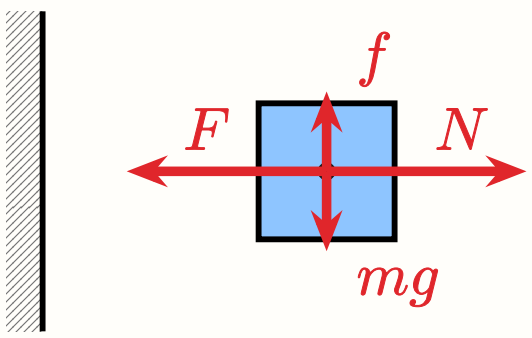
$$a = T\cos\alpha/m$$

α above the horizontal · valid while $T\sin\alpha < mg$

Example

$m = 5\,\text{kg}$, $T = 20\,\text{N}$, $\alpha = 30°$, $g = 10$: $N = 40\,\text{N}$, $a = 3.46\,\text{m s}^{-2}$.

3 Block pressed on a rough wall



- $N = F$; static $f = mg$ (up) while $mg \le \mu_s F$.
- $F = 50\,\text{N}$, $m = 2\,\text{kg}$, $\mu_s = 0.5$, $g = 10$: $f = 20\,\text{N}$, not $\mu_s N = 25\,\text{N}$ (arrow lengths to scale).

Tension, Strings and Pulleys

1 Ideal string and pulley

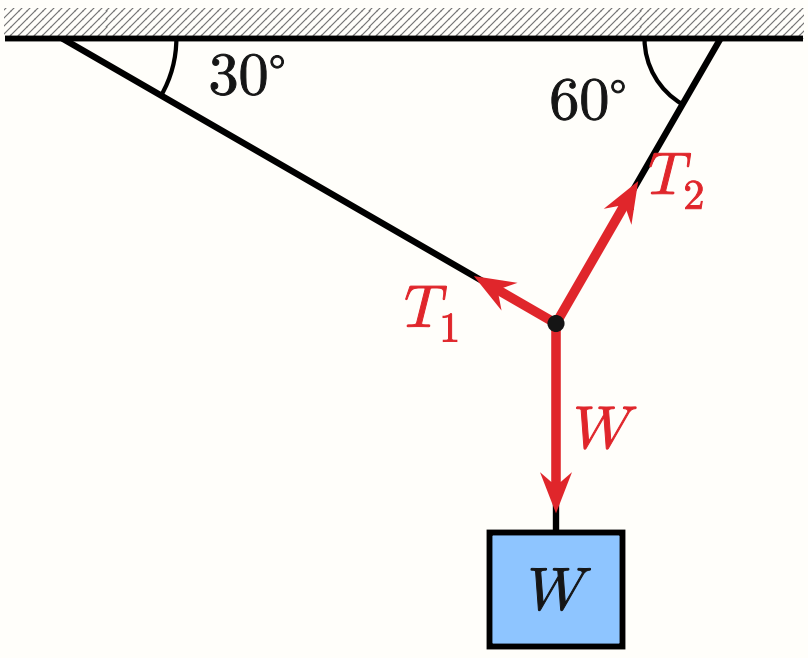
- Massless string: the same T at every point, even while accelerating.
- Inextensible: tied bodies share speed and acceleration along it.
- Ideal pulley (massless, frictionless): same T on both sides.
- A string only pulls: $T \geq 0$; $T = 0$ means slack.

2 Heavy (uniform) rope

$$T = \frac{x}{L} F \quad \text{(pulled)}, \qquad T = \frac{x}{L} Mg \quad \text{(hanging)}$$

mass M, length L · x: from the FREE end · pulled by F on a smooth floor; hanging at rest from its top

3 Knot in equilibrium: Lami's theorem



$$\frac{T_1}{\sin\alpha} = \frac{T_2}{\sin\beta} = \frac{W}{\sin\gamma}$$

three forces on a point, in equilibrium · each angle: between the OTHER two

Example

$W = 100\,\text{N}$: $\alpha = 150°$, $\beta = 120°$, $\gamma = 90°$; $T_1 = 50\,\text{N}$, $T_2 = 86.6\,\text{N}$ (to scale).

Spring Force

1 Hooke's law

$$\vec{F} = -k\vec{x}$$

x: change from the NATURAL length · k in $\mathrm{N\,m^{-1}}$ · within the elastic limit

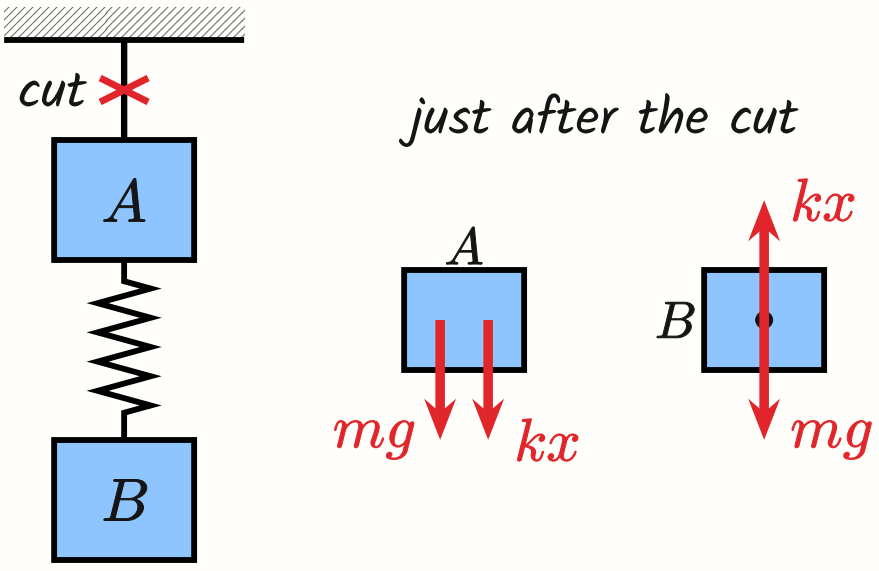
Example

$k = 200\,\mathrm{N\,m^{-1}}$, $2\,\mathrm{kg}$ hangs at rest, $g = 10\,\mathrm{m\,s^{-2}}$:
$x = mg/k = 0.1\,\mathrm{m}$.

2 Massless spring

- Same force at both ends: a balance pulled by $10\,\mathrm{N}$ at EACH end reads $10\,\mathrm{N}$.
- Series $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$; parallel $k = k_1 + k_2$; half a spring: $2k$.

3 A spring force cannot jump



- Equal masses; before: $kx = mg$.
- Cut: x needs time to change, so $kx = mg$ still.
- A: $2mg$ down, $a_A = 2g$.
- B: net 0, $a_B = 0$.

Golden points

★ String tension can jump; a spring's force cannot change instantly (while both its ends stay attached to bodies with mass).

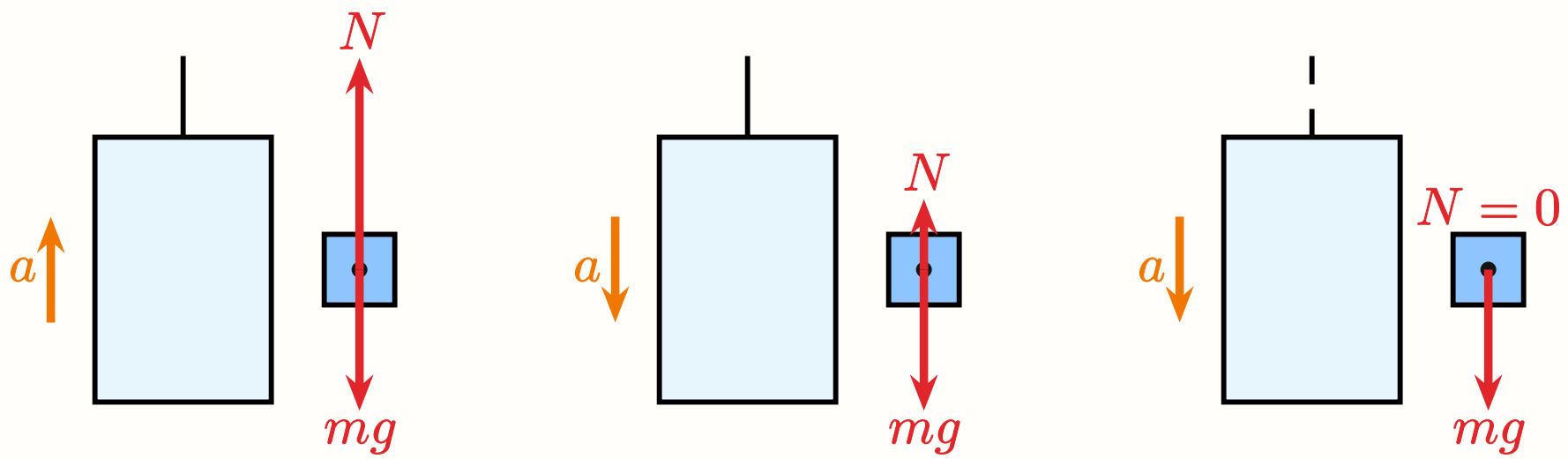
Apparent Weight in a Lift

1 Normal force on a person

$$N = m\,(g + a)$$

N: weighing-machine reading · a: lift's acceleration, $+$ up, from the ground frame · constant velocity: $N = mg$

2 Three cases (FBD of the person, to scale)



$a = g/2$ up | $a = g/2$ down | free fall, $a = g$

- a up (rising faster, or falling slower): $N > mg$, feels heavier (apparent weight $> mg$).
- a down (falling faster, or rising slower): $N < mg$.
- Free fall (cable cut): $N = 0$, "weightless"; m, mg unchanged.

Example

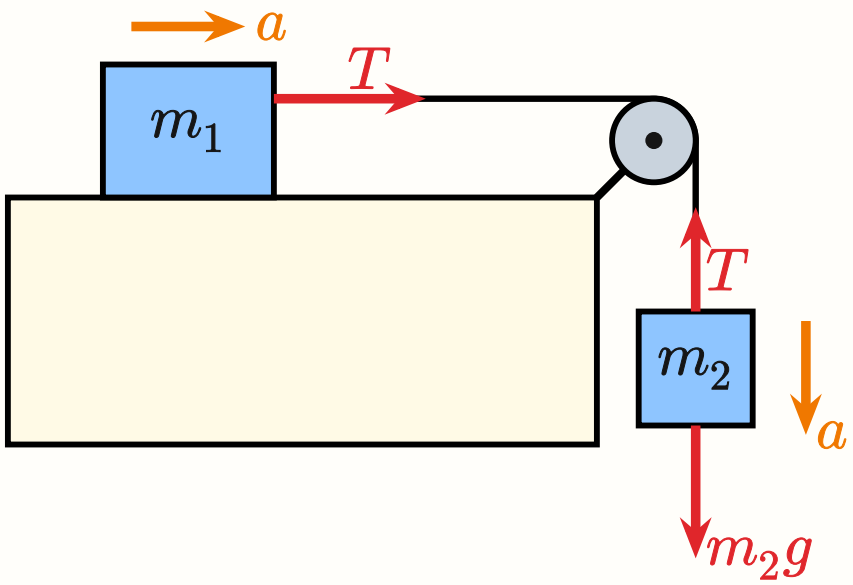
$60\,\mathrm{kg}$, $g = 10\,\mathrm{m\,s^{-2}}$: up at $2\,\mathrm{m\,s^{-2}}$, $N = 720\,\mathrm{N}$ (kg-scale shows $72\,\mathrm{kg}$); down at $2\,\mathrm{m\,s^{-2}}$, $N = 480\,\mathrm{N}$.

Golden points

★ In the lift $g_{eff} = g + a$ (a up): a dropped coin falls at $g + a$ relative to the floor.

Connected Bodies

1 Smooth table, hanging block



$$a = \frac{m_2\, g}{m_1 + m_2}$$

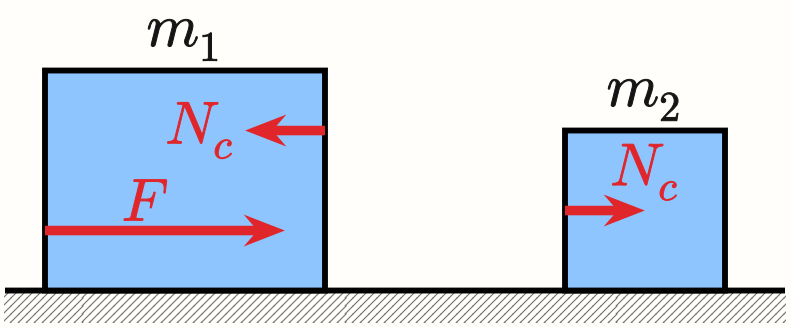
$$T = \frac{m_1 m_2\, g}{m_1 + m_2}$$

smooth table · ideal string and pulley

Example

$m_1 = 3$, $m_2 = 2\,\text{kg}$, $g = 10$: $a = 4\,\text{m s}^{-2}$, $T = 12\,\text{N}$ $(< m_2 g)$.

2 Blocks in contact, pushed by F



drawn apart; arrows to scale

$$a = \frac{F}{m_1 + m_2}$$

$$N_c = \frac{m_2\, F}{m_1 + m_2}$$

smooth floor · N_c: contact force

Example

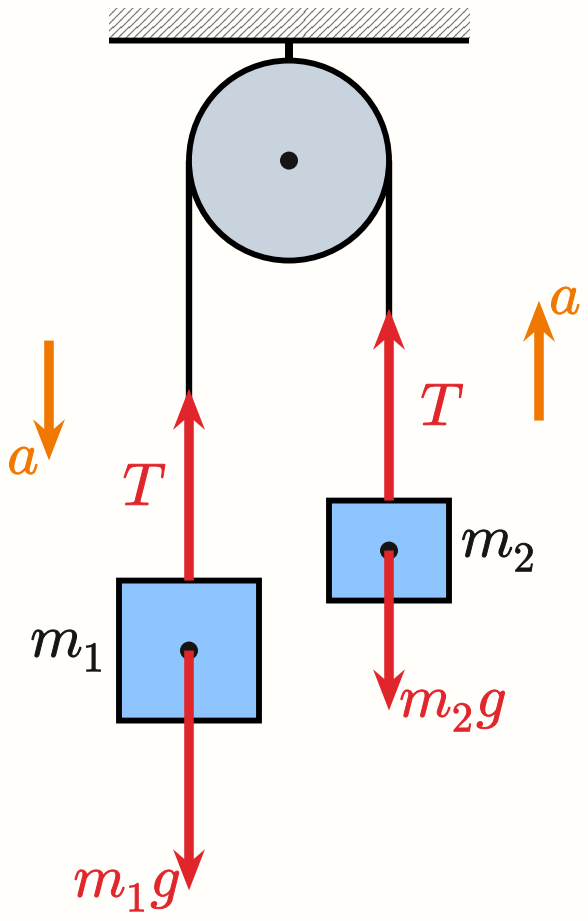
$F = 30\,\text{N}$, $m_1 = 4$, $m_2 = 2\,\text{kg}$: $a = 5\,\text{m s}^{-2}$, $N_c = 10\,\text{N}$; *push* m_2: $20\,\text{N}$.

Golden points

★ *Rough table: moves only if* $m_2 > \mu_s m_1$; *then* $a = (m_2 - \mu_k m_1)g/(m_1 + m_2)$.

Atwood Machine

1 Two blocks over a fixed pulley



$$a = \frac{(m_1 - m_2)\,g}{m_1 + m_2}$$

$$T = \frac{2m_1 m_2\,g}{m_1 + m_2}$$

$m_1 > m_2$ · *massless, inextensible string* · *ideal (massless, frictionless) pulley* · *ground frame*

- $m_1 g - T = m_1 a, \quad T - m_2 g = m_2 a.$
- *Support holds the pulley with* $2T$.
- *Force arrows drawn to scale.*

Example

$m_1 = 3\,\mathrm{kg}$, $m_2 = 2\,\mathrm{kg}$, $g = 10\,\mathrm{m\,s^{-2}}$, *from rest:* $a = 2\,\mathrm{m\,s^{-2}}$, $T = 24\,\mathrm{N}$, *support* $48\,\mathrm{N}$ *(<* *weight* $50\,\mathrm{N}$*); after* $2\,\mathrm{s}$*:* $v = 4\,\mathrm{m\,s^{-1}}$, *each has moved* $4\,\mathrm{m}$.

Golden points

★ T *lies between* $m_2 g$ *and* $m_1 g$. *Equal masses:* $a = 0$, $T = mg$.

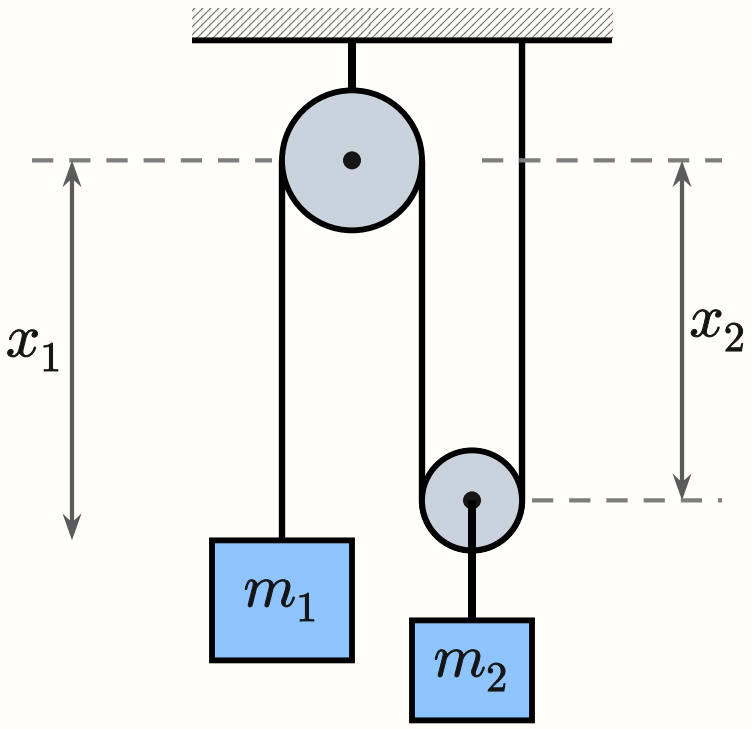
★ *In a lift accelerating UP at* a_0*: replace* g *by* $g + a_0$ *(*a *is then relative to the lift).*

Summary

$a = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$ • $T = \dfrac{2m_1 m_2 g}{m_1 + m_2}$ • *support* $2T$

Constraint Relations

1 Movable pulley



$$x_1 + 2x_2 = \text{const}$$

$$a_1 = -2\,a_2$$

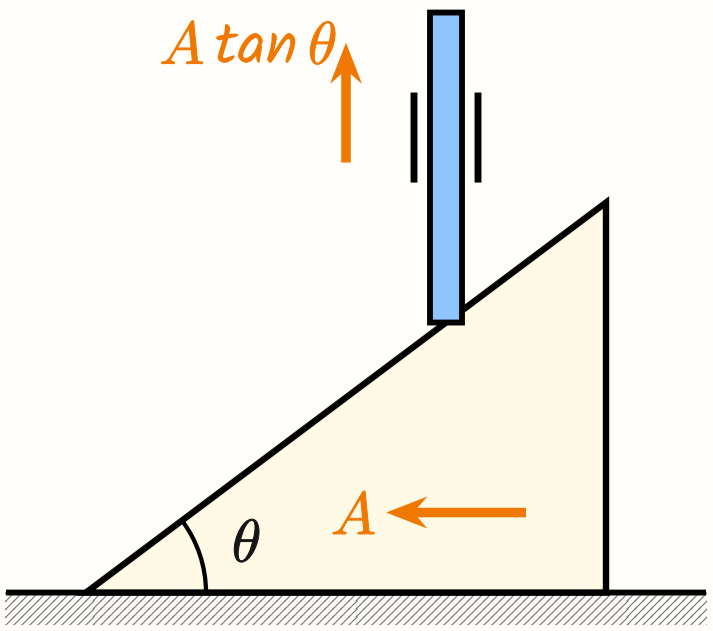
inextensible, massless string · ideal (massless, frictionless) pulleys · x_1, x_2: depths of m_1 and the lower axle

- *Massless movable pulley:* $2T$ *up.*
- m_1 *falls* $2\,\text{cm}$*:* m_2 *rises* $1\,\text{cm}$*.*

Example

$m_1 = m_2 = 1\,\text{kg}$, $g = 10$: m_1 *down* $4\,\text{m s}^{-2}$, m_2 *up* $2\,\text{m s}^{-2}$; $T = 6\,\text{N}$.

2 Wedge constraint



$$a_{rod} = A\tan\theta$$

vertical rod in guides, on a wedge moving left with A · in contact: equal acceleration components along the normal

Example

$\tan\theta = 3/4$, $A = 4\,\text{m s}^{-2}$: *rod rises at* $3\,\text{m s}^{-2}$.

Golden points

★ *Constraints: any masses; valid only for an inextensible string or a rigid contact, while the string stays taut / contact is kept.*

Pseudo Force

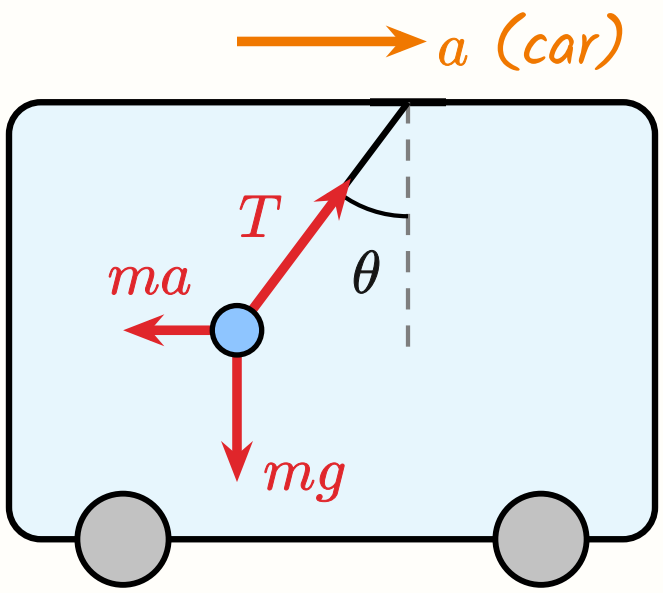
1 Non-inertial frame

$$\vec{F}_{pseudo} = -m\,\vec{a}_0$$

$\vec{a}_0$: frame's acceleration, from an inertial frame · on every mass · no agent, no partner

- Accelerating frame: real forces $-m\vec{a}_0 = m\vec{a}_{rel}$. Inertial frame: never add it.

2 Bob in an accelerating car



$$\tan\theta = \frac{a}{g}$$

$$T = m\sqrt{g^2 + a^2}$$

car frame, constant a on a level road; bob at rest in it · θ from the vertical; string hangs BACKWARD

Example

$a = 7.5$, $g = 10\,\mathrm{m\,s^{-2}}$: $\tan\theta = 3/4$ ($\theta \approx 37°$), $T = 1.25\,mg$ (to scale).

Golden points

★ Smooth wedge (angle θ) accelerated horizontally, sloping face in front: a block stays at rest on it (relative to the wedge) if $a = g\tan\theta$.

★ Centrifugal force $m\omega^2 r$ (outward): pseudo force seen only in a frame ROTATING at ω; not a reaction force.

Static and Kinetic Friction

1 Static friction (no slipping)

$$f_s \le \mu_s N$$

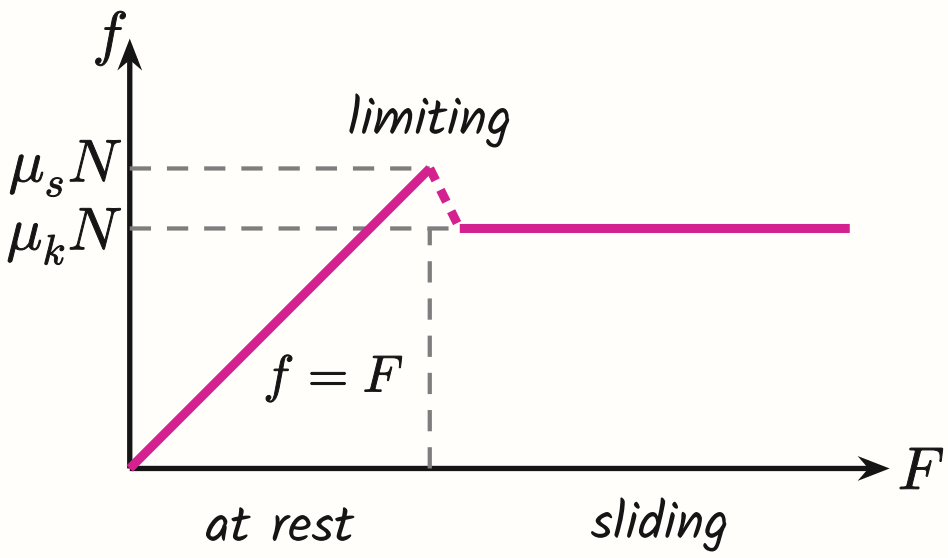
self-adjusting: just what stops slipping, up to $\mu_s N$ · opposes the TENDENCY of relative motion

2 Kinetic friction (sliding)

$$f_k = \mu_k N$$

only WHILE sliding · opposes relative motion · usually $\mu_k < \mu_s$ · nearly independent of speed and area

3 Friction vs applied force



- *Horizontal F; at rest $f = F$ (45°, equal scales).*
- *Peak $\mu_s N$; sliding: $\mu_k N$.*
- *Drawn for $\mu_k = 0.8\,\mu_s$.*

Example

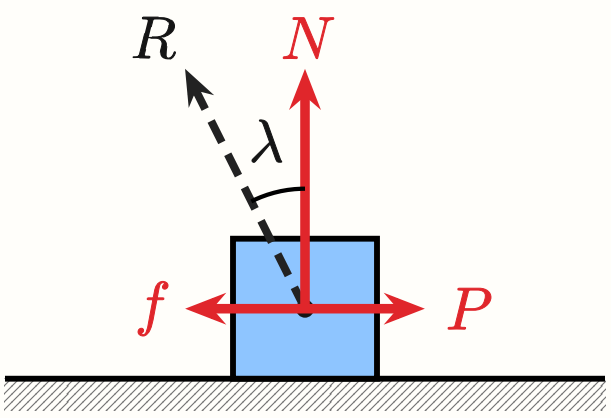
$m = 10\,\text{kg}$, level floor, horizontal F, $\mu_s = 0.5$, $\mu_k = 0.4$, $g = 10$: $F = 30\,\text{N}$ gives $f = 30\,\text{N}$ (rest); it starts at $50\,\text{N}$; at $F = 60\,\text{N}$: $f = 40\,\text{N}$, $a = 2\,\text{m s}^{-2}$.

Golden points

★ *Friction can be ALONG the motion: a box on an accelerating truck gets forward static friction while it does not slip.*

Angle of Friction and Repose

1 Angle of friction λ

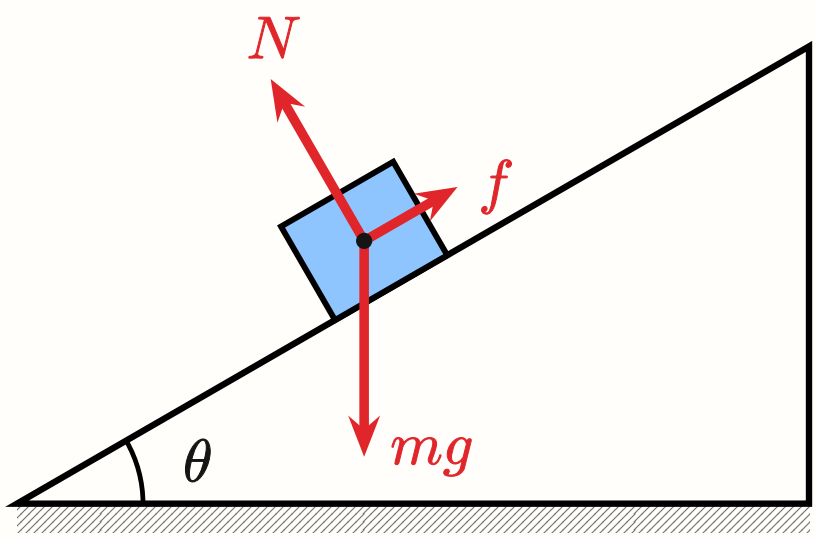


$$\tan\lambda = \frac{f_{max}}{N} = \mu_s$$

R: resultant of N and LIMITING friction · λ: angle from N to R

- *Horizontal pull P, drawn at the limit: $P = \mu_s N$, $\mu_s = 0.5$.*

2 Angle of repose θ_r



$$\tan\theta_r = \mu_s$$

incline tilted slowly until the block JUST slides · $\theta_r = \lambda$ · any mass

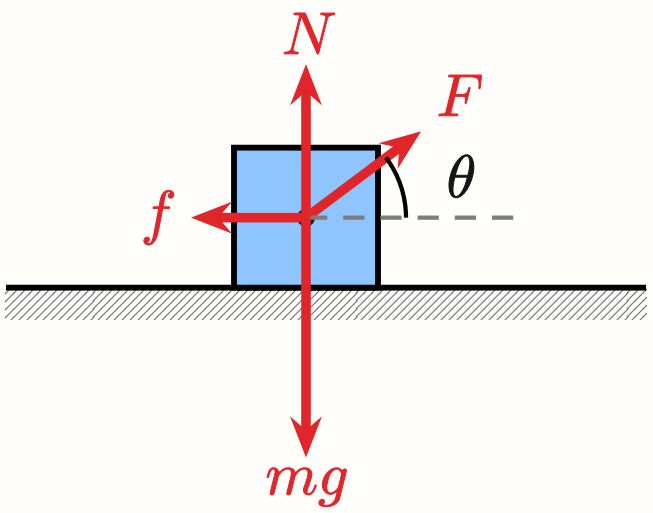
- *At rest: $N = mg\cos\theta$, $f = mg\sin\theta$ (self-adjusting); slides when $\tan\theta > \mu_s$.*
- *Drawn at $\theta = \theta_r = 30°$ to scale: $f = \mu_s N$ exactly.*

Example

$\mu_s = 1/\sqrt{3} \approx 0.577$: $\theta_r = 30°$. *At $20°$ it stays,* $f = mg\sin 20° < \mu_s N$.

Least Force to Move a Block

1 Pull at angle θ above the horizontal



$$F = \frac{\mu_s mg}{\cos\theta + \mu_s \sin\theta}$$

level floor · $N = mg - F\sin\theta$ · block just starts to move

$$F_{min} = \frac{\mu_s mg}{\sqrt{1+\mu_s^2}} \quad \text{at} \quad \tan\theta = \mu_s$$

best pull angle $\theta = \lambda$, the angle of friction · figure: the example's forces, to scale

2 Push at angle θ below the horizontal

$$F = \frac{\mu_s mg}{\cos\theta - \mu_s \sin\theta}$$

pushing presses it down: $N = mg + F\sin\theta$ · if $\tan\theta \geq 1/\mu_s$, no push can move it

Example

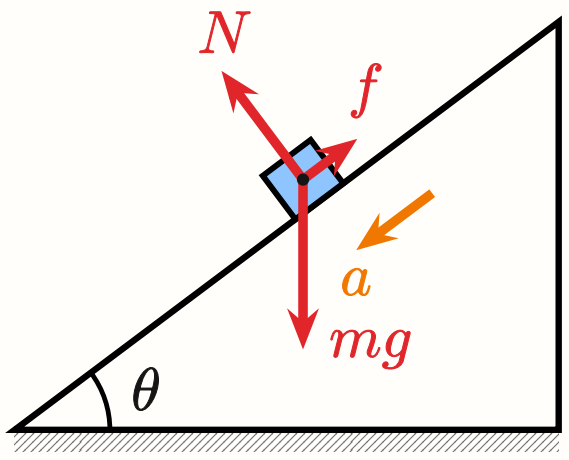
$m = 10\,\text{kg}$, $\mu_s = 0.75$, $g = 10$: *horizontal* $75\,\text{N}$; *pull at* $\tan^{-1}(3/4) \approx 37°$: $60\,\text{N}$ *(least); push at* $37°$: $\approx 214\,\text{N}$.

Golden points

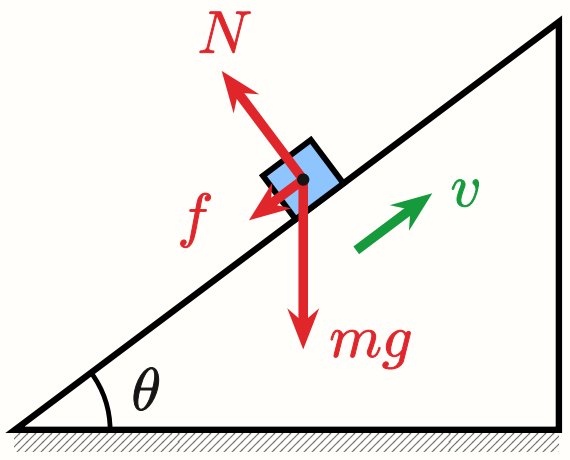
- ★ *At the same angle, pulling beats pushing (it reduces N).*
- ★ *To start a block UP a rough incline of angle α: least force $mg\sin(\alpha+\lambda)$, at λ above the incline.*

Block on a Rough Incline

1 Sliding down and moving up



sliding DOWN: f up the slope



moving UP: f down the slope

$$a_{down} = g(\sin\theta - \mu_k \cos\theta)$$

$$a_{up} = g(\sin\theta + \mu_k \cos\theta)$$

sliding down (from rest only if $\tan\theta > \mu_s$) · a_{up}: retardation going up

Example

$\theta = 37^\circ$ ($\sin = 0.6$, $\cos = 0.8$), $\mu_s = 0.6$, $\mu_k = 0.5$, $g = 10$: slides ($0.75 > 0.6$); $a_{down} = 2$, $a_{up} = 10\,\mathrm{m\,s^{-2}}$. To scale: $mg : N : f = 5 : 4 : 2$.

2 Held by a force F up the incline

$$mg(\sin\theta - \mu_s \cos\theta) \le F \le mg(\sin\theta + \mu_s \cos\theta)$$

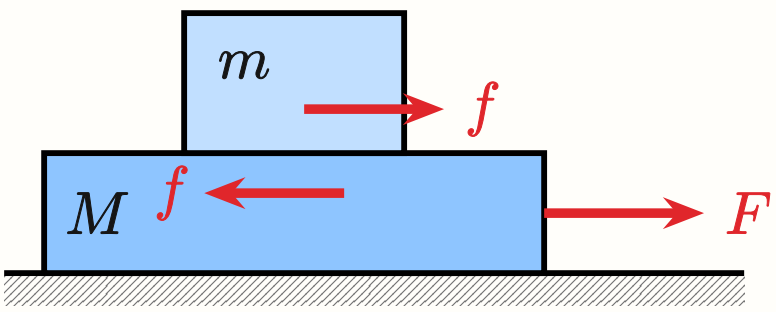
block at rest · same example, $m = 2\,\mathrm{kg}$: $2.4\,\mathrm{N} \le F \le 21.6\,\mathrm{N}$

Golden points

- ★ Thrown up: STAYS at the top if $\tan\theta \le \mu_s$; else slides back.
- ★ Slides down from rest in n times the smooth-incline time (same length): $\mu_k = (1 - \frac{1}{n^2})\tan\theta$.

Two-Block Friction

1 Force F on the lower block



smooth floor

- f on m: FORWARD; it is what accelerates m.
- f on M: backward (pair).
- Without slipping, m can reach at most $\mu_s g$.

$$F \le \mu_s(m+M)g: \quad a = \frac{F}{m+M}$$

$$F > \mu_s(m+M)g: \quad a_m = \mu_k g, \quad a_M = \frac{F-\mu_k mg}{M}$$

μ_s, μ_k: between the blocks · floor smooth · ground frame

Example

$m = 2\,\mathrm{kg}$, $M = 4\,\mathrm{kg}$, $\mu_s = 0.4$, $\mu_k = 0.3$, $g = 10$: *limit* $24\,\mathrm{N}$.
$F = 18\,\mathrm{N}$: *together*, $a = 3\,\mathrm{m\,s^{-2}}$, $f = 6\,\mathrm{N}$ $(< \mu_s mg = 8\,\mathrm{N})$.
$F = 36\,\mathrm{N}$: *slip*, $a_m = 3$, $a_M = 7.5\,\mathrm{m\,s^{-2}}$.

2 Force F on the upper block

no slip while $F \le \mu_s mg\dfrac{m+M}{M}$

floor smooth · then $a = F/(m+M)$ · after slipping: $a_m = (F-\mu_k mg)/m$, $a_M = \mu_k mg/M$

Golden points

★ Test "no slip" first: find the friction it needs, compare with $\mu_s N$.

Friction That Drives Motion

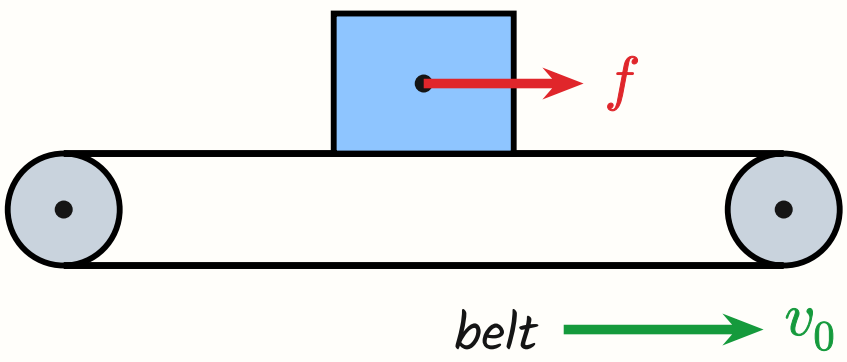
1 Walking and driving

- Walking: the foot pushes the ground back; static friction on the foot is FORWARD.
- Level road, all wheels driven, no skid: $a_{max} = \mu_s g$. Locked wheels skid to rest in $s = v^2/2\mu_k g$.

Example

$\mu_s = 0.6$, $g = 10$: $a_{max} = 6\,\mathrm{m\,s^{-2}}$. Skid from $20\,\mathrm{m\,s^{-1}}$, $\mu_k = 0.5$: $s = 40\,\mathrm{m}$.

2 Block placed on a moving belt



- Belt at constant v_0; block put on at rest.
- Kinetic $f = \mu_k m g$ FORWARD until the block reaches v_0.

$$t = \frac{v_0}{\mu_k g}, \qquad s_{slip} = \frac{v_0^2}{2\mu_k g}$$

level belt · s_{slip}: slip measured on the belt · then $f = 0$

Example

$v_0 = 2\,\mathrm{m\,s^{-1}}$, $\mu_k = 0.2$, $g = 10$: $t = 1\,\mathrm{s}$; ground frame: block $1\,\mathrm{m}$, belt $2\,\mathrm{m}$; slip $1\,\mathrm{m}$.

Golden points

★ Rolling friction $\ll$ sliding; lubricants reduce sliding friction.

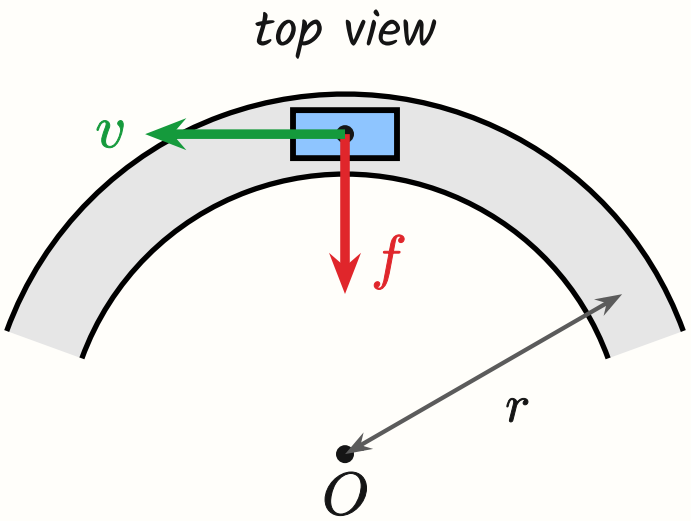
Circular Motion Dynamics

1 Centripetal force

$$F_c = \frac{mv^2}{r} = m\omega^2 r \quad \text{(towards the centre)}$$

NET radial force on a body in a circle of radius r · supplied by T, N, f or gravity · speed changing: also $F_t = m\,dv/dt$

2 Car on a level curve



$$v_{max} = \sqrt{\mu_s\, r\, g}$$

flat road · static friction towards O: $mv^2/r \le \mu_s mg$ · any mass

- $v > v_{max}$: *the car skids outward.*

Example

$r = 40\,\text{m}$, $\mu_s = 0.4$, $g = 10$:
$v_{max} = \sqrt{160} \approx 12.6\,\text{m s}^{-1} \approx 45.5\,\text{km h}^{-1}$.

Golden points

★ *No separate "mv^2/r" arrow in an FBD (inertial frame).*

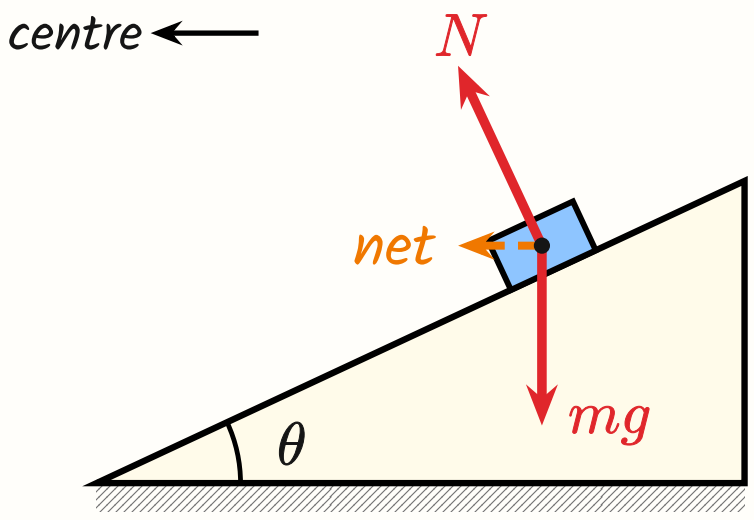
★ *If $\mu_s > d/2h$ it topples before skidding, at $v > \sqrt{g\,r\,d/2h}$ (d: track width, h: height of the CM).*

Summary

$F_c = mv^2/r$ • $v_{max} = \sqrt{\mu_s r g}$ • *not a new force*

Banking of Roads

1 No friction: optimum speed v_0



$$\tan\theta = \frac{v_0^2}{r\,g}$$

θ: bank angle · r: radius of the horizontal circle

- $N\cos\theta = mg$, $N\sin\theta = mv_0^2/r$.
- To scale, $\theta = 25°$: $N + mg$ is horizontal, towards the centre.

2 With friction μ_s

$$v_{max} = \sqrt{rg\,\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}}$$

$$v_{min} = \sqrt{rg\,\frac{\tan\theta - \mu_s}{1 + \mu_s\tan\theta}}$$

limiting static friction: DOWN the slope at v_{max}, UP at v_{min} · $v_{min} = 0$ if $\mu_s \geq \tan\theta$

Example

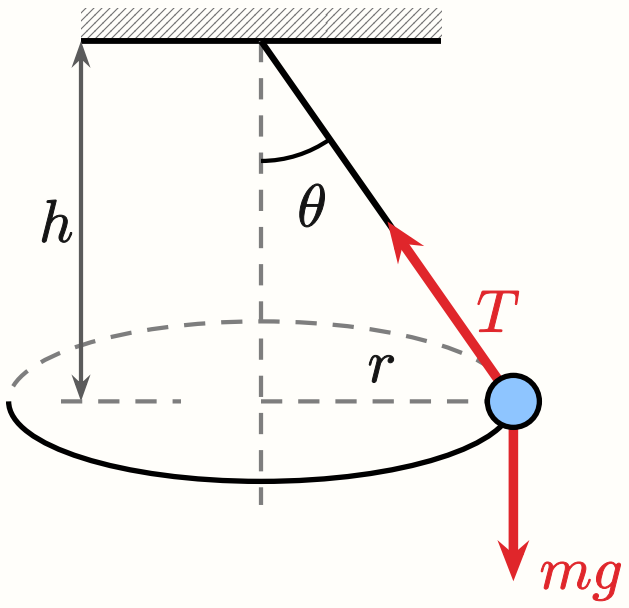
$r = 300\,\text{m}$, $\theta = 15°$, $\mu_s = 0.2$, $g = 10$: $v_0 \approx 28.4$, $v_{max} \approx 38.5$, $v_{min} \approx 13.9\,\text{m s}^{-1}$.

Golden points

★ Railway: outer rail raised h, gauge d: $\tan\theta \approx h/d$ (small θ).

Conical Pendulum and Death Well

1 Conical pendulum



$$\omega = \sqrt{\frac{g}{L\cos\theta}}, \qquad T = \frac{mg}{\cos\theta}$$

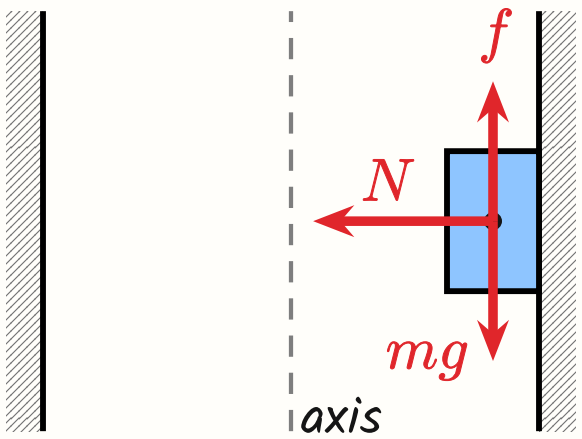
string length L, θ from the vertical · $r = L\sin\theta$, $h = L\cos\theta$ · horizontal circle

- $T\cos\theta = mg$, $T\sin\theta = m\omega^2 r$.
- Period $2\pi\sqrt{h/g}$. To scale at $35°$.

Example

$L = 1\,\mathrm{m}$, $\theta = 60°$, $g = 10$: $T = 2mg$, $\omega = \sqrt{20} \approx 4.47\,\mathrm{rad\,s^{-1}}$, period $\approx 1.40\,\mathrm{s}$. Always $T > mg$; $\theta = 90°$ is impossible.

2 Death well (vertical wall)



$$v_{min} = \sqrt{\frac{r\,g}{\mu_s}}$$

vertical wall · $N = mv^2/r$ towards the axis · static $f = mg$ needs $mg \le \mu_s N$ · r: radius

Example

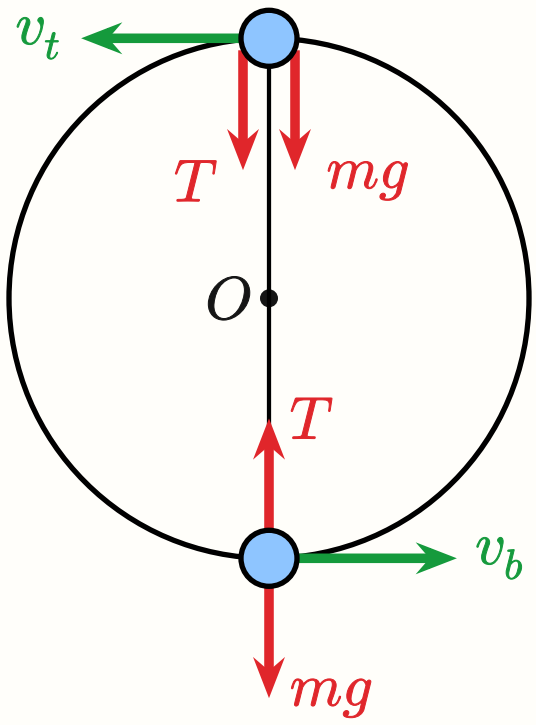
$r = 5\,\mathrm{m}$, $\mu_s = 0.5$, $g = 10$: $v_{min} = 10\,\mathrm{m\,s^{-1}}$.
Rotor: $\omega_{min} = \sqrt{g/\mu_s r}$.

Golden points

★ Well and rotor: the least speed does not depend on the rider's mass.

Vertical Circle

1 Bob on a string, radius R



$$v_t \geq \sqrt{gR}, \quad v_b \geq \sqrt{5gR}$$

$$T_b - T_t = 6mg$$

light inextensible string · only gravity does work ($T \perp v$): $v_b^2 = v_t^2 + 4gR$

- *Top:* $T + mg = mv_t^2/R$, $T \geq 0$.
- *At the limit:* $T_t = 0$, $T_b = 6mg$.

2 String vs rod: least speed at the bottom

- *String, or ball INSIDE a smooth loop (pulls / pushes one way only):* $v_b \geq \sqrt{5gR}$.
- *Light rod, or bead on a ring (push or pull):* $v_t \geq 0$, *so* $v_b \geq \sqrt{4gR} = 2\sqrt{gR}$.
- *String,* $\sqrt{2gR} < v_b < \sqrt{5gR}$*: goes slack above the horizontal diameter;* $v_b \leq \sqrt{2gR}$*: swings in the lower half.*

Example

$R = 1\,\mathrm{m}$, $g = 10$: *string* $v_b \geq \sqrt{50} \approx 7.07\,\mathrm{m\,s^{-1}}$ ($v_t \geq \sqrt{10} \approx 3.16$); *rod* $v_b \geq \sqrt{40} \approx 6.32\,\mathrm{m\,s^{-1}}$.

Golden points

★ *Car on top of a hump (radius r):* $N = mg - mv^2/r$; *it loses contact ($N = 0$) at* $v = \sqrt{gr}$.

Common Traps

1 Exam traps

Avoid these

- ✘ Cancelling an action–reaction pair (different bodies).
- ✘ $N = mg$ always (not on an incline, in a lift, with a slanted force).
- ✘ $f = \mu_s N$ for a body at rest (static f only goes UP TO $\mu_s N$).
- ✘ Friction always against the body's motion (it opposes RELATIVE slipping).
- ✘ Same T on both sides of a pulley with mass or friction.
- ✘ Pseudo force in an inertial frame; "mv^2/r" as an extra force.
- ✘ $F = ma$ with changing mass (use $F = dp/dt$).
- ✘ $\sqrt{5gR}$ for a rod (it is $\sqrt{4gR}$).
- ✘ Weightless = no gravity (only $N = 0$; mg still acts).
- ✘ Momentum conserved in a collision with a large external impulse (e.g. a fixed wall).

2 Solving recipe

Steps

- Frame: inertial, or add $-m\vec{a}_0$.
- FBD of each body; axes along $\vec{a}$.
- $\sum F = ma$ per axis; then constraints.
- Friction: assume no slip, check $f \le \mu_s N$.

Summary

which frame? • which body? • slipping or not?